MUKESH KUMAR DHARANI

HSE Engineer | Mining & Energy Sector

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PROFESSIONAL SUMMARY

HSE Engineer with 6+ years of independent field experience across underground coal mining, coal gasification, and industrial research. Analyzed a 1.2-million man-hour LTI-free record— consistently exceeding safety targets in high-consequence environments. Proficient in ISO 45001, ISO 14001, HIRA, JSA, RCA/CAPA, permit-to-work systems, gas monitoring, and emergency response planning, with a strong track record of building safety cultures from the ground up in greenfield and operational settings. Currently completing an M.E. in Georesource & Geoenergy Engineering at Politecnico di Torino, with NEBOSH IGC in progress. Available for HSE roles in the GCC, Europe, and Pakistan.

PROFESSIONAL EXPERIENCE

HSE Site Engineer ***August 2022 – July 2025***

Abbas-R Construction Company, Hyderabad Sindh, Pakistan

- Drove a **45% reduction in workplace incidents** by designing and deploying site-specific HSE plans, risk assessments, and JSAs across high-risk construction activities in full compliance with **ISO 45001** and **ISO 14001**.
- Sustained **1.2 million man-hours LTI-free** by leading 300+ toolbox talks, safety inductions, and workforce training programs, elevating PPE compliance from **70% to 98%** site-wide.
- Delivered **95%+ compliance scores** across all internal and external audits by maintaining rigorous HSE documentation, incident investigation records, and corrective action registers.
- Reduced near-miss incidents by **35%** through behavioral safety programs and 20+ emergency mock drills, measurably improving site emergency response readiness.
- Cut site waste by **25%** and ensured full regulatory compliance on dust, noise, and waste disposal by implementing ISO 14001-aligned environmental controls and leading workforce sustainability campaigns.

HSE Site Engineer ***March 2019 – June 2022***

Sindh Lakhra Coal Mining Company (SLCMC) — Lakhra Coal Mine, Sindh, Pakistan

- **Implemented** a site-wide HSE management system aligned with ISO 45001 & ISO 14001, achieving 100% regulatory compliance across underground and surface coal mining operations serving a workforce of 300+.
- **Led** quarterly HIRA and JSA reviews for high-risk activities including drilling, blasting, confined space entry, and coal extraction — reducing near-miss incidents by 40% within 18 months.
- **Engineered** a real-time gas monitoring programme (CH_4, CO, H_2S) integrated with mine ventilation controls, reducing atmospheric hazard alerts by 35% and achieving zero gas-related LTIs over 3 consecutive years.
- **Conducted** 60+ safety audits and inspections annually, identifying and closing non-conformities within defined CAPA timelines, maintaining an average closure rate of 95% within 30 days.
- **Delivered** monthly toolbox talks and safety inductions to 100+ workers and contractors, driving PPE compliance from 72% to 98% within 12 months of programme rollout.

HSE Assistant ***January 2017 – June 2018***

UCG Project Thar — Government of Sindh, Coal & Energy Development Department, Thar, Sindh, Pakistan

- **Established** HSE policies, PTW systems, and emergency response procedures from the ground-up for a Greenfield underground coal gasification (UCG) project — achieving zero LTIs across all high-risk commissioning and operational phases.
- **Executed** comprehensive HIRA and JSA for drilling, well installation, and syngas operations — identifying 25+ critical hazard scenarios and implementing controls that reduced task-level risk ratings by 55%.
- **Deployed** continuous multi-gas monitoring systems (CO, CH_4, H_2S) across active gasification zones, enabling real-time hazard detection and reducing unplanned emergency responses by 40% over the project lifecycle.
- **Assessed** safety inductions, enclosed space rescue drills, and gas leak response exercises to 50+ workers and contractors, establishing a safety culture in a newly mobilized, high-consequence work environment.
- **Designed** and implemented a restricted space entry management programme for well and borehole operations, reducing exposure time by 25% through pre-entry atmospheric testing and standby rescue protocols.

EDUCATION

M.E. — Georesource and Geoenergy Engineering *Politecnico di Torino, Turin, Italy* *Sep 2025 – Present*
B.E. — Mining Engineering *Mehran University of Engineering & Technology, Pakistan* *Jan 2013 – Dec 2016*

CERTIFICATIONS

- Health, Safety & Environmental (HSE) Engineering Specialization — Khalifa University / Coursera, Mar 2026
- Risk Management and Industrial Safety — Khalifa University / Coursera, Mar 2026

TECHNICAL & SOFT SKILLS

ISO 45001, ISO 14001, Hazard Identification and Risk Assessment (HIRA), Job Safety Analysis (JSA), Root Cause Analysis (RCA), Corrective and Preventive Action (CAPA), Incident Investigation, Safety Audits, Site Inspections, Emergency Response Planning, Permit-to-Work Systems, PPE Management, Gas Detection, Ventilation Monitoring, Fire Prevention Systems, Underground Mining Safety, Environmental Compliance, Regulatory Compliance, KPI Reporting, HSE Documentation, Toolbox Talks, Safety Training, Risk Management, Engineering, Microsoft Office, Microsoft Excel, Leadership, Communication, Decision Making, Team Collaboration, Stakeholder Management, Initiative, Independent Work, Conflict Resolution, Cross-Cultural Communication.

LANGUAGES

English (Professional Proficiency); **Urdu / Sindhi** (Native); **Italian** (Intermediate — B1).